



CSCI 523 - Parallel Systems and GPU Programming Параллель жүйелер және GPU бағдарламалау Параллельные системы и программирование на GPU

General Course Information

School: School of Engineering

Course Type: Major Core Course

Semester in which the course is offered: Fall & Spring

Course instructor(s): Ben Tyler

Talgat Manglayev

Course Description - English

This course is intended for master's students interested in the efficient use of modern parallel systems. Topics such as parallel computer architecture, programming models, memory hierarchy, parallel program design and parallel programming tools for multi-core systems and general-purpose graphics processing units (GPGPUs) will be covered. This comprehensive overview will equip students with a broad understanding of the key approaches in heterogeneous programming. Students will engage in practical, hands-on projects utilizing real-world scientific models to tackle contemporary challenges and applications, including physics, chemistry, biology and trustworthy machine learning. In the second part, the course covers the most common and current GPU parallel programming techniques with lab-based programming assignments using CUDA API with C/C++ and Python.

Course Description - Kazakh

Бұл курс заманауи параллель жүйелерді тиімді пайдалануға қызығушылық танытатын магистранттарға арналған. Параллельді компьютер архитектурасы, бағдарламалау модельдері, жад иерархиясы, параллельді бағдарламаларды жобалау және көп ядролы жүйелер мен жалпы мақсаттағы графикалық процессорлар (GPGPU) үшін параллель бағдарламалау құралдары сияқты тақырыптар қарастырылады. Бұл жан-жақты шолу студенттерді гетерогенді бағдарламалаудағы негізгі тәсілдер туралы кең түсінікпен қамтамасыз етеді. Студенттер физика, химия,

биология және сенімді машиналық оқытуды қоса алғанда, заманауи қиындықтар мен қолданбаларды шешу үшін индустриядағы нақты ғылыми үлгілерді пайдаланатын қолма-қол, тәжірибелік жобалармен айналысады. Курстың екінші бөлімінде студенттер C/C++ және Python көмегімен CUDA API арқылы зертханалық бағдарламалау тапсырмалары бар GPU параллельді бағдарламалаудың ең кең таралған және ағымдағы әдістерін үйренеді.

Course Description - Russian

Этот курс предназначен для студентов магистратуры, интересующихся эффективным использованием современных параллельных систем. Будут рассмотрены такие темы, как архитектура параллельных компьютеров, модели программирования, иерархия памяти, проектирование параллельных программ и инструменты параллельного программирования для многоядерных систем и графических процессоров общего назначения (GPGPU). Этот всесторонний обзор обеспечит студентов широким пониманием ключевых подходов в гетерогенном программировании. Студенты будут участвовать в практических проектах, используя реальные научные модели для решения современных задач и приложений, включая физику, химию, биологию и надежное машинное обучение. Во второй части курса рассматриваются наиболее распространенные и актуальные техники параллельного программирования на GPU с лабораторными заданиями по программированию, использующими API CUDA с C/C++ и Python.

Course Requisites

Conditions of Enrollment

Course Aims

- Introduce students to concepts, hardware architectures and software programming models of parallel systems
- Develop knowledge and understanding of parallel programming technologies
- Develop hands-on experience skills in designing and implementing simple parallel programs
- Develop knowledge and understanding of running programs on GPU
- Ensure students can apply the learned methodologies to tackle contemporary challenges in science fields

Course Learning Outcomes (CLOs)

Course Learning Outcomes	Graduate Attributes
Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded.
Define concepts related to CPU and GPU hardware architectures from mobile multi-coreto server/cluster many-core parallel systems.	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded.

Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks	GA1 - Possess an in-depth and sophisticated understanding of their domain of study. GA2 - Be intellectually agile, curious, creative and open-minded. GA6 - Be cultured and tolerant citizens of the world.
Design and implement parallel programs using the parallel programming techniques and CUDA	GA 1. Possess in-depth knowledge and understanding of their discipline, and are engaged in the pursuit of new knowledge GA 3. Are problem-solvers, able to adapt and apply critical and evidence-based thinking to meet evolving challenges GA 7. Leverage information and digital literacy, and artificial intelligence (AI), for their academic studies, development and careers
Integrate current research findings, methodologies, and advancements into heterogeneous computing	GA3 - Be thoughtful decision makers who know how to involve others. GA5 - Be fluent and nuanced communicators across languages and cultures.
Solve parallel programming problems using C++ multithreading	GA 1. Possess in-depth knowledge and understanding of their discipline, and are engaged in the pursuit of new knowledge GA 3. Are problem-solvers, able to adapt and apply critical and evidence-based thinking to meet evolving challenges GA 7. Leverage information and digital literacy, and artificial intelligence (AI), for their academic studies, development and careers

Mapping of Major Core and Elective Courses

Program: Master of Science in Computer Science (Active)

Course Learning Outcomes	Program Learning Outcomes
Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	2. Assess complex technical problems, and design and implement solutions in the form of devices or software (practical skills).
Define concepts related to CPU and GPU hardware architectures from mobile multi-coreto server/cluster many-core parallel systems.	2. Assess complex technical problems, and design and implement solutions in the form of devices or software (practical skills).
Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks	2. Assess complex technical problems, and design and implement solutions in the form of devices or software (practical skills). 6. Acknowledge the need for ongoing personal and professional development, so as to continuously acquire the knowledge and skills necessary to remain informed and effective (professional development).
Design and implement parallel programs using the parallel programming techniques and CUDA	2. Assess complex technical problems, and design and implement solutions in the form of devices or software (practical skills). 6. Acknowledge the need for ongoing personal and professional development, so as to continuously acquire the knowledge and skills necessary to remain informed and effective (professional development).
Integrate current research findings, methodologies, and advancements into heterogeneous computing	2. Assess complex technical problems, and design and implement solutions in the form of devices or software (practical skills).

Solve parallel programming problems using C++ multithreading	2. Assess complex technical problems, and design and implement solutions in the form of devices or software (practical skills).
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ECTS and Student Workload

Credits: 6

Contact Hours: 45

Self-Study Hours: 105

Total Hours: 150

Assessment Methods

Assessment Methods	Course Learning Outcomes (Reference #)
Lab and Homework Assignments	1, 2, 3, 4, 5, 6
Quizzes (Two)	1, 2, 3, 4, 5, 6
Midterm Exams (Two)	1, 2, 3, 4, 5, 6

Assessment Methods Weighting

Assessment Methods	Weighting
Lab and Homework Assignments	30%
Quizzes (Two)	30%
Midterm Exams (Two)	40%
Total weighting	100%

Course Outline

Session Number	Theme (Key Topic)	Course Learning Outcomes (Reference #)
1	Course Overview. Introduction to parallel systems	1, 2
2	Fundamentals of Accelerated Computing with CUDA C/C++	2, 3, 4, 5, 6
3	Fundamentals of Accelerated Computing with Modern CUDA C++	2, 3, 4, 5, 6

4	Fundamentals of Accelerated Computing with Modern CUDA C++	2, 3, 4, 5, 6
5	Optimize CUDA programs using Nsight profiling tools.	1, 2, 3, 4, 5
6	CUDA C++ features. Reduction algorithm implementation variations using CUDA C/C++.	1, 2, 3, 4
7	Review and Midterm	1, 2, 3, 4, 5, 6
8	Fundamentals of accelerated computing with CUDA python. Introduction to CUDA with Numba. Custom CUDA Kernels in Python with Numba.	1, 2, 3, 4
9	Effective Use of the Memory Subsystem. Nsight Analysis System: Build Custom Python Analysis Scripts.	1, 2, 3, 4
10	Fundamentals of Accelerated Data Science.	3, 4
11	Fundamentals of Accelerated Data Science.	3, 4
12	Fundamentals of Deep Learning with Python calling CUDA functions.	1, 2, 3, 4
13	Fundamentals of Deep Learning with Python calling CUDA functions.	1, 2, 3, 4
14	Research issues on recent parallel systems. Summary and Course Review	1, 2, 3, 4

Learning and Teaching Methods

Course Learning Outcomes	Learning and Teaching Methods
Articulate the differences of diverse parallel systems with the connection of parallel programming tools/APIs	Laboratories Homeworks; Quizzes; Exams
Define concepts related to CPU and GPU hardware architectures from mobile multi-core to server/cluster many-core parallel systems.	Laboratories Lectures Homeworks; Quizzes; Exams
Critically assess the performance of various CUDA applications, utilizing both quantitative metrics and qualitative insights, to identify the bottlenecks	Lectures Laboratories Homeworks; Quizzes; Exams
Design and implement parallel programs using the parallel programming techniques and CUDA	Laboratories Lectures Homeworks; Quizzes; Exams
Integrate current research findings, methodologies, and advancements into heterogeneous computing	Laboratories Homeworks; Quizzes; Exams

Solve parallel programming problems using C++ multithreading	Laboratories Lectures Homeworks, Quizzes, Exams;
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Laboratory Content

During labs, students are asked to solve tasks relevant to a new topic presented to them. The tasks have prepared code and students need to solve the remaining part of the task. Students are invited one by one to the teacher’s PC, which is projected to the class, to solve a task. The other students may either work on the task independently or follow the selected student’s solution. After completing the task, the student explains their approach to the class.

The students complete approximately one NVIDIA workshop every two weeks, where any remaining tasks are completed at home. In this way, students can earn a certificate of completion from NVIDIA, which they need to upload to receive a grade for their lab.

E-Learning

The course is not designed as an online course, but will make use of some online materials.

Online platform learning.nvidia.com Deep Learning Institute of NVIDIA. The online courses are step-by-step tutorials which have tasks at the end of each course.

Grading

Graduate

Letter Grade	Grade (quality points)	Grade Descriptors
A	4.00	Excellent; exceeds the highest standards in the assignment or course
A-	3.67	Excellent; meets the highest standards for the assignment or course
B+	3.33	Very good; meets high standards for the assignment or course
B	3.00	Good; meets most of the standards for the assignment or course
B-	2.67	More than adequate; shows some reasonable command of the material
C+	2.33	Acceptable; meets basic standards for the assignment or course
C	2.00	Acceptable; meets some of the basic standards for the assignment or course
C-	1.67	Acceptable; while falling short of meeting basic standards in several ways
F	0	Failing; very poor performance

Key Books and Learning Materials

Textbook				
#	Author(s):	Title	Publisher	Year
1	Wen-mei W. Hwu, David B. Kirk and Izzat El Hajj	Programming Massively Parallel Processors: A Hands-on Approach	Morgan Kaufmann	2026
2	Jason Sanders, Kandrot Edward and Jack Dongarra	CUDA by Example	Upper Saddle River, N.J: Addison-Wesley	2011
3	John L. Hennessy and David A. Patterson	Computer Organization and Design 4th Edition (Ch. 7)	MK Publications	
4	John L. Hennessy and David A. Patterson	Computer Architecture: A Quantitative Approach 5th Edition (Ch. 3 and Ch. 4)	MK Publications	

Other Learning Materials

CUDA C Programming Guide (E-resources)

Academic Integrity Statement

[Student Code of Conduct and Disciplinary Procedures](#)

While general guidelines regarding academic integrity are published in the NU Student Handbook and by the school, for this course you must adhere to the following rules:

- You are not allowed to distribute or indirectly make your work available to others, even if you are not intending them to copy it -- this is still considered academic misconduct.
- While it is okay to talk through high-level concepts and ideas with your fellow students and to study with them, you are not allowed to give or receive direct help from others on graded assignments.
- Using code that you find online is strictly forbidden, as is using code generation tools such as ChatGPT, MS Copilot, Github Copilot, etc.
- Any appearance of cheating or inappropriate copying should be avoided, as such instances will be investigated and reported to the school.
- You may only get help on graded work from designated people—the instructors or TAs for the course. If you are struggling with something, by all means, please seek help from them.

In the event that academic misconduct such as plagiarism or cheating is discovered, the student will receive no credit for the work, and the event reported to your school. Egregious cases, or a second offense, can result in failure of the course and potential suspension or expulsion from the university.

When a student suspects that another student has violated the academic honesty policy, a report should be made to the appropriate faculty member.

Course Expectations

The Nazarbayev University Special Learning Needs Committee (SLNC) is committed to creating an equitable and inclusive education environment for all students. If you have a qualified special learning need (physical, cognitive, socio-emotional, and psychological), please contact the SLNC as early as possible to ensure you receive the fullest support available.

For more information, please contact us at SLNC@nu.edu.kz and follow this [link](#) to the Accommodation Request Instructions.

Course Expectations

Attendance

Missing classes and habitual tardiness will have a negative effect on your grade, both directly and indirectly. You are also responsible for any announcements made during the class period, so be sure to ask your instructor, TAs, or classmates for any info that you may have missed if you did not attend.

Electronic Resources

Labs will be conducted in one of our hybrid computer labs, which are designed to accommodate the full range of course activities. The necessary programming tools are installed on the classroom lab computers, and are available online for free download to your own computer. You are expected to check your Nazarbayev University e-mail on a daily basis for updates and announcements about the course.

Assignment Submission and Late Policy

Assignments must be submitted by the announced due date and time, as directed by the instructor. Some assignments may need to be submitted in the form of physical hard-copy in class, generally within the first five minutes of the start of the class period. Other assignments will need to be submitted digitally to Moodle. In case Moodle does not work, assignments need to be submitted by email to your instructor AND teaching assistant by the deadline date and time. In general, there is no late policy; if you submit an assignment after it is due, you get zero points for your assignment. In cases of illness or family emergency, you must inform your instructor immediately if you believe you will not be able to submit your assignment on time. In such cases, an exception may be made at the discretion of your instructor.

Classroom Behavior

You are expected to act respectfully towards your fellow classmates, TAs, and instructors inside and outside of the classroom. We have a limited amount of time to cover a lot of material this semester, so you need to pay attention in class, and do your in-class work when it is assigned. Talking on your phone, texting, chatting online, browsing Facebook or other social media sites, and talking excessively with your neighbors about non-class related stuff are just a few examples of behavior that is not acceptable, and will negatively impact your grade.

Grade Policy. The passing grade for the course is C- and above. As per CS department policy, students with failing marks within 5% of the passing threshold do not have the option of receiving an Additional Assessment Task at the end of the semester to increase their grade.

Supporting Documents

Course Approval Date

School Learning and Teaching Committee Approval:

Course Origination Date: 2026-06-16

Effective from: